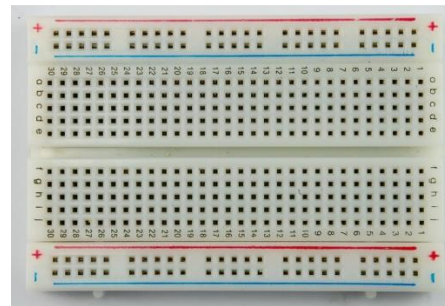
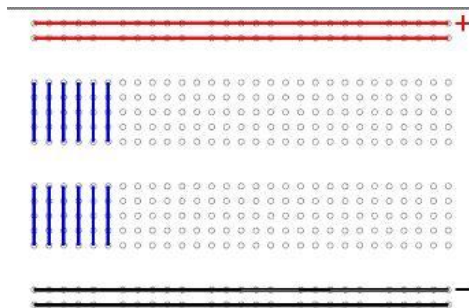


Basic Electronics

What is Bread Board :

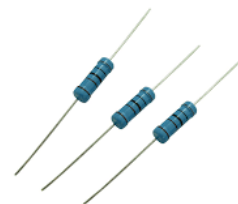
A **breadboard** is a platform you can **use** to build and test electronic circuits, usually without having to do any soldering. Certain parts of the **breadboard** are wired together so that electricity can flow from component to component in orderly row



What is Resistor :

A **resistor** is a passive two-terminal electrical component that implements electrical resistance as a circuit element. In electronic circuits, **resistors** are used to reduce current flow, adjust signal levels, to divide voltages, bias active elements, and terminate transmission lines, among other uses. Resistance is measured in ohms

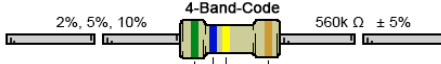
Resistors are made out of either a carbon, metal, or metal-oxide film.



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Resistor Color Coding :

Standard Resistor Values and Color Components and wires are coded with colors to identify their value and function. The colors **brown, red, green, blue, and violet** are used as tolerance codes on 5-band resistors only. All 5-band resistors use a colored tolerance band.



COLOR	1 ST BAND	2 ND BAND	3 RD BAND	MULTIPLIER	TOLERANCE
Black	0	0	0	1Ω	
Brown	1	1	1	10Ω	± 1% (F)
Red	2	2	2	100Ω	± 2% (G)
Orange	3	3	3	1KΩ	
Yellow	4	4	4	10KΩ	
Green	5	5	5	100KΩ	± 0.5% (D)
Blue	6	6	6	1MΩ	± 0.25% (C)
Violet	7	7	7	10MΩ	± 0.10% (B)
Grey	8	8	8	100MΩ	± 0.05%
White	9	9	9	1GΩ	
Gold				0.1Ω	± 5% (J)
Silver				0.01Ω	± 10% (K)

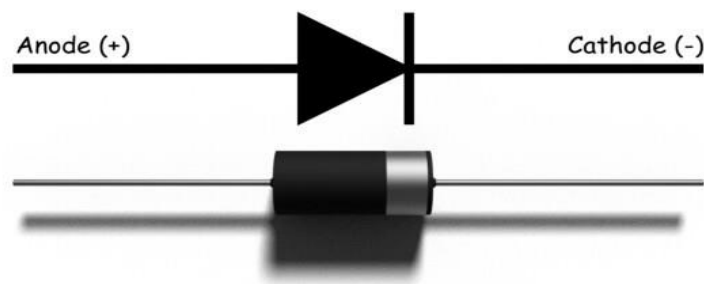


Color	Numeric Value	Multiplier	Tolerance	Temperature coefficient
BLACK	0	1Ω		250
BROWN	1	10 Ω	±1%	100
RED	2	100 Ω	±2%	50
ORANGE	3	1K Ω		15
YELLOW	4	10 Ω		25
GREEN	5	100 Ω	±0.5%	20
BLUE	6	1M Ω	±0.25%	10
VIOLET	7		±0.1%	5
GREY	8			1
WHITE	9			
GOLD			±5%	
SILVER			±10%	

What is Diode :

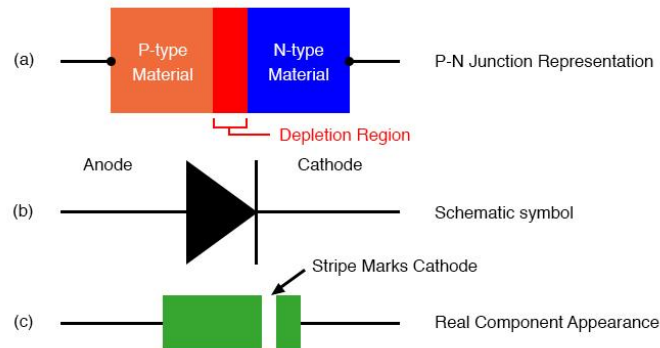
A **diode** is defined as a two-terminal electronic component that only conducts current in one direction (so long as it is operated within a specified voltage level). An ideal **diode** will have zero resistance in one direction, and infinite resistance in the reverse direction

Capacitance is measured in units called farads



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Diodes are made with semiconductor materials such as silicon, germanium,



What is capacitor

A **capacitor** is a device that stores electrical energy in an electric field. It is a passive electronic component with two terminals. The effect of a **capacitor** is known as capacitance.

Capacitance is measured in units called farads

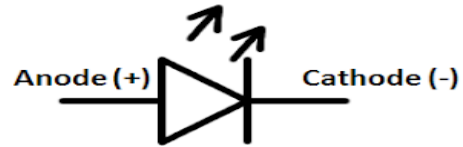
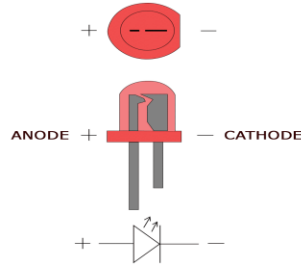


Capacitor are made of a conductive material: aluminum, tantalum, silver, or other metals.

What is LED (Light emitting diode) :

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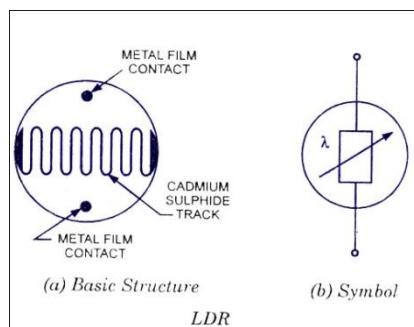
LEDs, which are also known as Light Emitting Diodes, burn light 90 percent more efficiently than incandescent bulbs. LEDs work by passing an electrical current through a microchip, which causes tiny light emitting diodes to illuminate. **LED** lights can last for years before they will need to be changed.



LED manufacture are gallium arsenide (GaAs), gallium phosphide (GaP), or gallium arsenide phosphide (GaAsP).

LDRs (light-dependent resistors)

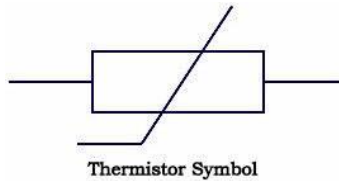
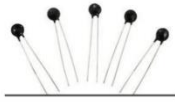
LDRs (light-dependent resistors) are used to detect light levels, for example, in automatic security lights. Their resistance decreases as the light intensity increases: in the dark and at low light levels, the resistance of an **LDR** is high and little current can flow through it



What is Thermistor:

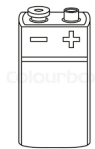
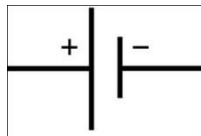
A **thermistor** is a resistance thermometer, or a resistor whose resistance is dependent on temperature. ... This type of **thermistor** is used the most. A PTC **thermistor works** a little differently. When temperature increases, the resistance increases, and when temperature decreases, resistance decreases.

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What is Battery :

A **battery** converts chemical energy into electrical energy by a chemical reaction. Usually the chemicals are kept inside the **battery**. It is **used in** a circuit to power other components. A **battery** generally consists of an anode, a cathode, and an electrolyte



What is Switch :

In **electrical** engineering, a **switch** is an **electrical** component that **can** disconnect or connect the conducting path in an **electrical** circuit, interrupting the electric current or diverting it from one conductor to another.



What is Motor :

An electric **motor** is an electrical machine that converts electrical energy into mechanical energy.



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Testing of components

Aim of the experiment:

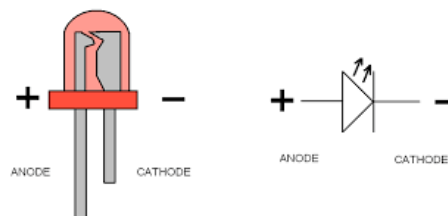
To test the given led with Battery

Gadgetry:

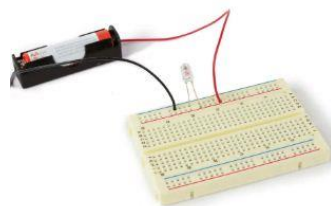
- LEDs
- Resistor (220,330ohm)
- Breadboard
- Jumper cables
- Battery(9v)
- Switch

Theory:

A Light Emitting Diode (LED) generates light in a semi-conductor material, which is an electronic component. Using the right materials, a diode may produce visible light of various wavelengths. LED technology is a sensible alternative to traditional light sources in many applications.



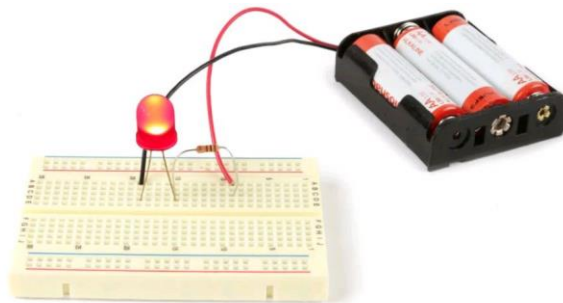
Do it yourself :



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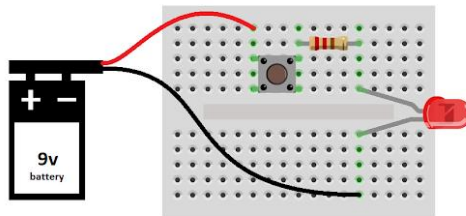
With Without Resistor First I had to decide what power source to use and which LED to light up. too much voltage is bad and can burn out the LED instantaneously.

Resistor with LED :



It was just a coincidence that I bought an LED that was 1.7V and that it ended up working being able to be powered by my 1.5V power supply without the use of a resistor. For this second setup I decided to use the same LED,

LED With Switch :



In **electrical** engineering, a **switch** is an **electrical** component that **can** disconnect or connect the conducting path in an **electrical** circuit

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Aim of the experiment:

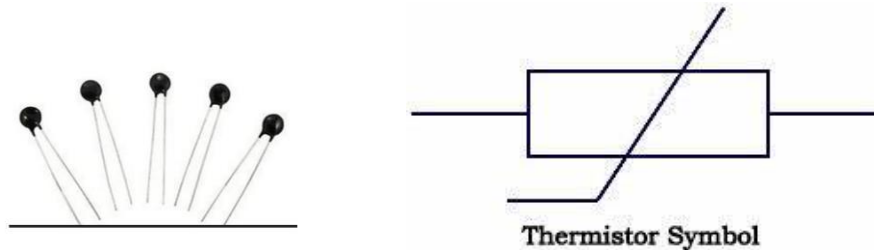
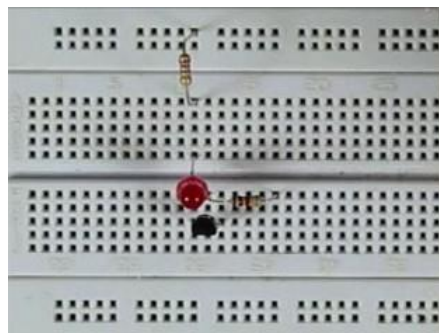
To test the given thermistor using Battery

Gadgetry:

- Thermistor
- 10K, 330ohms resistor
- Breadboard
- Jumper cables
- Battery(9v)

Theory:

A **thermistor** is a resistance thermometer, or a resistor whose resistance is dependent on temperature. ... This type of **thermistor** is used the most. A PTC **thermistor works** a little differently. When temperature increases, the resistance increases, and when temperature decreases, resistance decreases.

**Do it yourself:**

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Different Uses for Thermistors :

- Microwave. For those who have used a microwave, you have used a thermistor. ...
- Circuit Protector. If you have a power supply or surge protector in your home or office then you are also using a thermistor. ...
- Automotive. Cars, trucks, and buses all use thermistors. ...
- Digital Thermometers. ...
- Rechargeable Batteries.

Aim of the experiment:

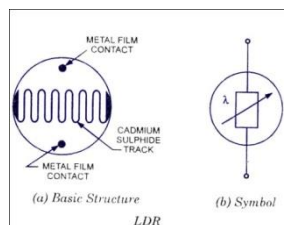
To test the given Light Dependent Resistors using Battery

Gadgetry:

- Light Dependent Resistors(LDRs)
- 10K, 330ohms resistor
- Breadboard
- Jumper cables
- Battery(9v)

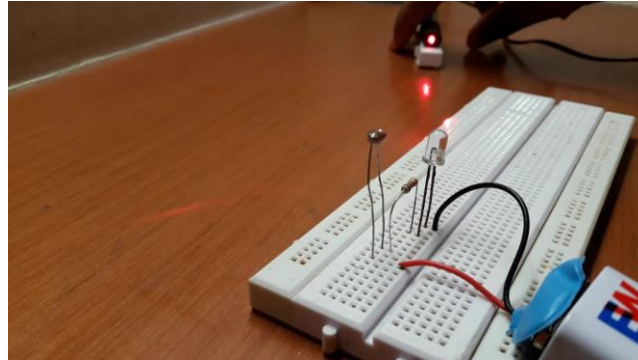
Theory:**LDRs (light-dependent resistors) :**

LDRs (light-dependent resistors) are used to detect light levels, for example, in automatic security lights. Their resistance decreases as the light intensity increases: in the dark and at low light levels, the resistance of an **LDR** is high and little current can flow through it



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Do it yourself:



Advantage of LDRs :

- Automatic street light system
- Automatic LED Emergency Light circuit

Aim of the experiment:

To test the given IR Sensor using Battery

Gadgetry:

- IR Sensor
- Breadboard
- Jumper cables
- Battery(9v)

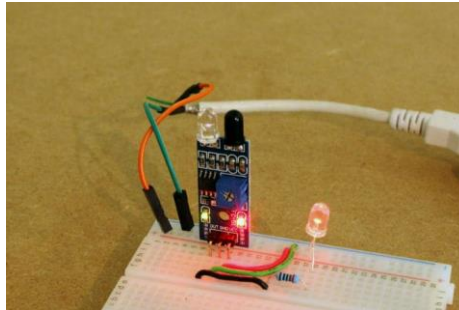
Theory:

It is an electronic device that measures and detects infrared radiation in its surrounding environment.

An object can be detected with an infrared system consisting of an infrared transmitter and a receiver. The IR transmitter sends an infrared signal that, in case of a reflecting surface (e.g. white color),

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When the surface is absorbent (e.g. black color) the IR signal isn't reflected and the object cannot be detected by the sensor.

Do it yourself:**Applications of IR Sensors :**

- Gas detectors.
- Water analysis.
- Anesthesiology testing.
- Petroleum exploration.
- Rail safety.

Aim of the experiment:

To test the given PIR Sensor using Battery

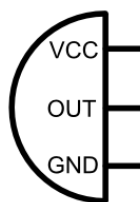
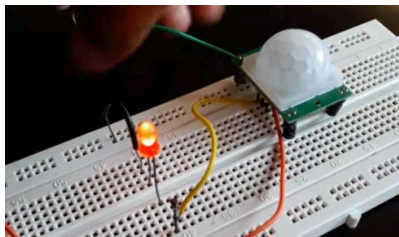
Gadgetry:

- PIR Sensor
- LED
- Bread board
- Jumper cables
- Battery(9v)

Theory :

A passive infrared **sensor** (**PIR sensor**) is an electronic **sensor** that measures infrared (IR) light radiating from objects in its field of view.

They **work** entirely by detecting infrared radiation (radiant heat) emitted by or reflected from objects.

Do it yourself:

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Application of PIR Sensors :

- Automatic switching operation of outdoor lights,
- Lift lobby,
- Automatic switching operation of garden lights based on the presence of a human being,
- For covered parking area,
- Automatic door operating system in shopping malls,

TESTING OF MOTORS

Aim of the experiment:

To test the given Motor using Battery

Gadgetry:

- DC BO Motor
- Bread board
- Jumper cables
- Battery(9v)

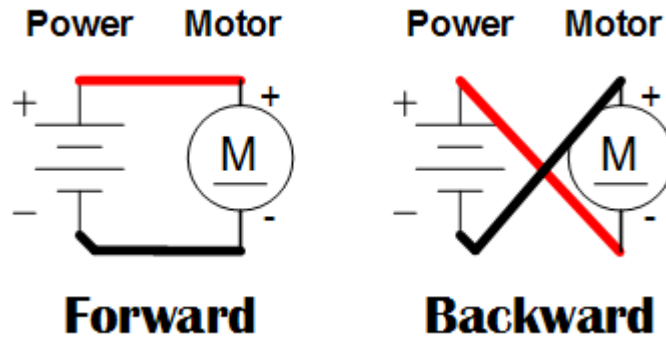
Theory :

An electric **motor** is an electrical machine that converts electrical energy into mechanical energy.

Most electric **motors** operate through the interaction between the **motor's** magnetic field and electric current in a wire winding to generate force in the form of rotation of a shaft.

Do it yourself:

Battery Without switch :



Battery With Switch:

